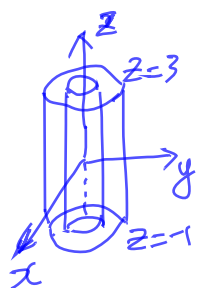


Quiz 8

Instructions: There are totally 4 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Using the cylindrical coordinates to compute the triple integral $\iiint_W z \sqrt{x^2 + y^2} \, dx dy dz$ where W is the solid that lies between the cylinders $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$, and between the planes $z = -1$ and $z = 3$.



$$W^* = \{(r, \theta, z) \mid 1 \leq r \leq 3, 0 \leq \theta \leq 2\pi, -1 \leq z \leq 3\}$$

$$\iiint_W z \sqrt{x^2 + y^2} \, dx dy dz = \iiint_{W^*} z \sqrt{x^2 + y^2} \cdot r \, dr d\theta dz$$

$$= \int_{-1}^3 \int_0^{2\pi} \int_1^3 z r \cdot r \, dr d\theta dz$$

$$= \int_{-1}^3 \int_0^{2\pi} z \left[\frac{r^3}{3} \right]_{r=1}^{r=3} d\theta dz = \int_{-1}^3 \int_0^{2\pi} \frac{26}{3} z \, d\theta dz$$

$$= \int_{-1}^3 \frac{26}{3} \cdot 2\pi z \, dz = \int_{-1}^3 \frac{52\pi}{3} z \, dz$$

$$= \frac{26}{3} [z^2]_{z=-1}^{z=3} = \frac{26}{3} (9-1) = \frac{208}{3}$$

Problem 2. (5 points) Find the volume of the solid defined in spherical coordinates by $W^* = \{(\rho, \phi, \theta) \mid 1 \leq \rho \leq 3, 0 \leq \theta \leq \pi/2, \pi/3 \leq \phi \leq \pi/2\}$.

$$\text{Volume} = \iiint_{W^*} \rho^2 \sin \phi \, d\rho d\phi d\theta$$

$$= \int_0^{\pi/2} \int_{\pi/3}^{\pi/2} \int_1^3 \rho^2 \sin \phi \, d\rho d\phi d\theta$$

$$= \int_0^{\pi/2} \int_{\pi/3}^{\pi/2} \sin \phi \left[\frac{\rho^3}{3} \right]_{\rho=1}^{\rho=3} d\phi d\theta$$

$$= \int_0^{\pi/2} \int_{\pi/3}^{\pi/2} \frac{26}{3} \sin \phi \, d\phi d\theta = \int_0^{\pi/2} \frac{26}{3} [-\cos \phi]_{\phi=\pi/3}^{\phi=\pi/2} d\theta$$

$$= \int_0^{\pi/2} \frac{13}{3} d\theta = \frac{13}{3} \cdot \frac{\pi}{2} = \frac{13}{6} \pi$$

Problem 3. (5 points) Evaluate the scalar line integral $\int_C x \sin y \, ds$ where C is the line segment from $(0, 3)$ to $(4, 6)$.

$$C: \mathbf{x}(t) = (0, 3) + t[(4, 6) - (0, 3)] = (0, 3) + t(4, 3) \\ = (4t, 3+3t), \quad 0 \leq t \leq 1$$

$$\Rightarrow \mathbf{x}'(t) = (4, 3), \quad \|\mathbf{x}'(t)\| = \sqrt{4^2 + 3^2} = 5$$

$$\begin{aligned} \int_C x \sin y \, ds &= \int_0^1 (4t) \sin(3+3t) \|\mathbf{x}'(t)\| \, dt \\ &= \int_0^1 20t \sin(3+3t) \, dt \\ &= 20t \left(-\frac{1}{3} \cos(3+3t)\right) \Big|_{t=0}^{t=1} - \int_0^1 20 \left(-\frac{1}{3} \cos(3+3t)\right) \, dt \\ &= -\frac{20}{3} \cos 6 + \int_0^1 \frac{20}{3} \cos(3+3t) \, dt \\ &= -\frac{20}{3} \cos 6 + \left[\frac{20}{9} \sin(3+3t)\right]_{t=0}^{t=1} = -\frac{20}{3} \cos 6 + \frac{20}{9} (\sin 6 - \sin 3) \end{aligned}$$

Problem 4. (5 points) Evaluate the vector line integral $\int_{\mathbf{x}} \mathbf{F} \cdot d\mathbf{s}$ where $\mathbf{F}(x, y, z) = (x + y)\mathbf{i} + (y - z)\mathbf{j} + z^2\mathbf{k}$ and the path $\mathbf{x}(t) = 2t^2\mathbf{i} + t^3\mathbf{j} + t^2\mathbf{k}$, $0 \leq t \leq 1$.

$$\mathbf{x}(t) = (2t^2, t^3, t^2), \quad 0 \leq t \leq 1$$

$$\Rightarrow \mathbf{x}'(t) = (4t, 3t^2, 2t), \quad 0 \leq t \leq 1$$

$$\begin{aligned} \int_{\mathbf{x}} \mathbf{F} \cdot d\mathbf{s} &= \int_0^1 (x+y, y-z, z^2) \cdot \mathbf{x}'(t) \, dt \\ &= \int_0^1 (2t^2+t^3, t^3-t^2, t^4) \cdot (4t, 3t^2, 2t) \, dt \\ &= \int_0^1 (2t^2+t^3) \cdot (4t) + (t^3-t^2) \cdot (3t^2) + t^4 \cdot (2t) \, dt \\ &= \int_0^1 8t^3 + 4t^4 + 3t^5 - 3t^4 + 2t^5 \, dt \\ &= \int_0^1 5t^3 + t^4 + 8t^5 \, dt \\ &= \left[\frac{5}{4}t^4 + \frac{1}{5}t^5 + 2t^6\right]_{t=0}^{t=1} \\ &= \frac{5}{4} + \frac{1}{5} + 2 = \frac{91}{20} \end{aligned}$$